DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-2026
MONDAY TEST

CLASS: X | **FULL MARKS: 40**
SUBJECT: CHEMISTRY | **DATE: 28/04/2025**

General Instructions:

- **The paper consists of three printed pages.**
- **Read the questions very carefully. Answers should be to the point.**
- **Question numbers should be copied carefully while answering the questions.**

SECTION A (20 MARKS)

Attempt all questions

Question 1

Choose the correct answers from the options given below: **[8]**

a) Some elements from the Periodic Table are given below:
Carbon, aluminium, fluorine, oxygen
How many element(s) has/ have electronegativity greater than that of nitrogen?
(i) 1
(ii) 2
(iii) 3
(iv) 4

b) An aqueous solution of lead nitrate forms a colourless solution on addition of:
(i) Excess sodium hydroxide
(ii) Drop wise sodium hydroxide
(iii) Excess ammonium hydroxide
(iv) Drop wise ammonium hydroxide

c) Assertion (A): Electronegativity of chlorine is higher than that of fluorine.
Reason (R) : Chlorine is placed just below fluorine in the same group in the periodic table.
(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, and R is the not the correct explanation of A.
(iii) A is false but R is true.
(iv) A is true but R is false.

d) If the empirical mass of the formula PQ_2 is 10, and the vapour density is 15, then the molecular formula will be:
(i) PQ_2
(ii) P_3Q_2
(iii) P_6Q_3
(iv) P_3Q_6

e) Choose the incorrect answer:
(i) The covalent bonds are said to be polar when the shared pair of electrons are not at equal distance between the two bonded atoms.
(ii) The bond formed between two atoms with slight difference in electronegativity is non polar.
(iii) The bond formed between two atoms with low electronegativity difference is ionic.
(iv) Polar covalent compounds ionise in water.

f) Assertion (A): Ions like Mg^{2+}, Na^+, F^- and O^{2-} are isoelectronic species.
Reason (R) : All isoelectronic species contain different number of electrons.
(i) A is true and R is the correct explanation of R.
(ii) Both A and R ar e true, but R is not the correct explanation of A.
(iii) Both A and R are false.
(iv) R is false, but A is the true explanation.

g) Elements A, B and C have electronic configuration as (2, 8, 3), (2, 8, 5) and (2, 8, 7) respectively. Which of the following holds true?
   (i) B has the least electron affinity.
   (ii) A has the highest electron affinity.
   (iii) C has the highest electron affinity.
   (iv) B has higher electronegativity than C.

h) The compound which does not contain - one single, a double or a triple bond in its molecule.
   (i) Nitrogen
   (ii) Chlorine
   (iii) Methane
   (iv) Oxygen

Question 2

a) The electronic configuration of three elements X, Y and Z are given below: [3]

Elements	Electronic configuration
X	2, 2
Y	2 8, 2
Z	2, 8, 8, 2

Arrange the elements X, Y and Z as per the instructions given below:
   (i) Decreasing order of atomic size
   (ii) Increasing order of metallic character
   (iii) Increasing order of ionisation potential

b) Draw the electron dot diagram for the following compounds given below: [3]
   (i) Calcium oxide
   (ii) Hydronium ion
   (iii) Nitrogen molecule

c) Calculate the empirical formula of the compound containing 37.6% sodium, 23.1% silicon, 39.3% oxygen. (Atomic mass: Na= 23, Si= 28, O=16) [3]

d) The following table shows the tests a student performed on three different aqueous solutions X and Y. Based on the observations provided, identify the cations in X and Y. [2]

Chemical Test		Observation	Cation
(i)	To a solution 'X' NH_4OH is added in minimum quantity first and then in excess	A gelatinous white ppt. is formed which dissolves in excess to form a clear solution	?
(ii)	To a solution 'Y' NH_4OH is added in minimum quantity first and then in excess	A pale blue ppt. is formed which dissolves in excess to form a clear inky blue solution	?

e) Define mole. [1]

SECTION B (20 MARKS)

Attempt all questions

Question 3

a) Calculate the percentage of nitrogen in aluminium nitride. [2]
   [At. mass: Al = 27, N= 14, O=16]

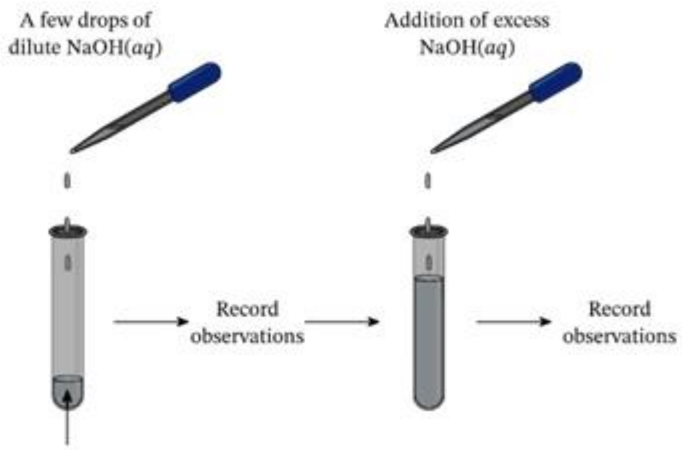
b) Inert gases have zero electron affinity. Account for the statement. [1]

c) Calculate the volume of oxygen required to burn completely a mixture of 22.4dm^3 of methane and 11.2dm^3 of hydrogen. All volumes are measured at STP. (1 dm^3=1litre) [2]

Question 4

a) Arun derived the molecular formula of a compound 'A' which has the following percentage composition by mass: C= 26.7%, O=71.1%, H= 2.2%. He found the relative molecular mass of 'A' to be 90. Find the molecular formula derived by him.
[At. Mass: C=12, H=1, O=16] **[4]**

b) Study the diagram given below and write your observation. **[1]**



Lead nitrate solution

Question 5

a) Two non metals combine with each other by sharing of electrons to form a compound 'X'. **[3]**
   (i) What type of chemical bond is present in 'X'?
   (ii) Will it be a good conductor of electricity or not? Give reason.

b) Solve the following numericals: **[2]**
   (i) A gas of mass 32g has a volume of 20litre at STP. Calculate the molecular weight of the gas.
   (ii) Calculate the number of moles present in 160g of sodium hydroxide.
   [At. Mass: Na=23, O=16]

Question 6:

Atomic number of element 'A' is 11 and it forms an ionic compound with 'B'. **[5]**

a) Which of the following atomic numbers will match B?
   (i) 14 (ii) 10 (iii) 9

b) Write the formula of the compound formed between A and B. Draw the electron dot structure of the compound formed between A and B.

c) What is the name given to the members of the group to which B belongs to?

d) Element ____ is an alkali metal.